

Voltar Company manufactures and sells a specialized cordless telephone for high electromagnetic radiation environments. The company's contribution format income statement for the most recent year is given below:

	Total	Per Unit	Percent of Sales
Sales (20,000 units)	\$1,200,000	\$60	100%
Variable expenses	900,000	45	? %
Contribution margin	300,000	\$15	? %
Fixed expenses	240,000		
Net operating income	\$ 60,000		

Management is anxious to increase the company's profit and has asked for an analysis of a number of items.

Required:

1. Compute the company's CM ratio and variable expense ratio.
2. Compute the company's break-even point in both units and sales dollars. Use the equation method.
3. Assume that sales increase by \$400,000 next year. If cost behavior patterns remain unchanged, by how much will the company's net operating income increase? Use the CM ratio to compute your answer.
4. Refer to the original data. Assume that next year management wants the company to earn a profit of at least \$90,000. How many units will have to be sold to meet this target profit?
5. Refer to the original data. Compute the company's margin of safety in both dollar and percentage form.

6.
 - a. Compute the company's degree of operating leverage at the present level of sales.
 - b. Assume that through a more intense effort by the sales staff, the company's sales increase by 8% next year. By what percentage would you expect net operating income to increase? Use the degree of operating leverage to obtain your answer.
 - c. Verify your answer to (b) by preparing a new contribution format income statement showing an 8% increase in sales.
7. In an effort to increase sales and profits, management is considering the use of a higher-quality speaker. The higher-quality speaker would increase variable costs by \$3 per unit, but management could eliminate one quality inspector who is paid a salary of \$30,000 per year. The sales manager estimates that the higher-quality speaker would increase annual sales by at least 20%.
 - a. Assuming that changes are made as described above, prepare a projected contribution format income statement for next year. Show data on a total, per unit, and percentage basis.
 - b. Compute the company's new break-even point in both units and dollars of sales. Use the formula method.
 - c. Would you recommend that the changes be made?

Solution to Review Problem

1.

$$\text{CM ratio} = \frac{\text{Unit contribution margin}}{\text{Unit selling price}} = \frac{\$15}{\$60} = 25\%$$

$$\text{Variable expense ratio} = \frac{\text{Variable expense}}{\text{Selling price}} = \frac{\$45}{\$60} = 75\%$$

2.

$$\begin{aligned} \text{Profit} &= \text{Unit CM} \times Q - \text{Fixed expenses} \\ \$0 &= (\$60 - \$45) \times Q - \$240,000 \\ \$15Q &= \$240,000 \\ Q &= \$240,000 \div \$15 \\ Q &= 16,000 \text{ units; or at } \$60 \text{ per unit, } \$960,000 \end{aligned}$$

3.

Increase in sales	\$400,000
Multiply by the CM ratio	× 25%
Expected increase in contribution margin	\$100,000

Because the fixed expenses are not expected to change, net operating income will increase by the entire \$100,000 increase in contribution margin computed above.

4. Equation method:

$$\begin{aligned} \text{Profit} &= \text{Unit CM} \times Q - \text{Fixed expenses} \\ \$90,000 &= (\$60 - \$45) \times Q - \$240,000 \\ \$15Q &= \$90,000 + \$240,000 \\ Q &= \$330,000 \div \$15 \\ Q &= 22,000 \text{ units} \end{aligned}$$

Formula method:

$$\begin{aligned} \text{Unit sales to attain} &= \frac{\text{Target profit} + \text{Fixed expenses}}{\text{Contribution margin per unit}} = \frac{\$90,000 + \$240,000}{\$15 \text{ per unit}} = 22,000 \text{ units} \\ \text{the target profit} & \end{aligned}$$

$$\begin{aligned}
 5. \quad \text{Margin of safety in dollars} &= \text{Total sales} - \text{Break-even sales} \\
 &= \$1,200,000 - \$960,000 = \$240,000 \\
 \text{Margin of safety percentage} &= \frac{\text{Margin of safety in dollars}}{\text{Total sales}} = \frac{\$240,000}{\$1,200,000} = 20\%
 \end{aligned}$$

$$\begin{aligned}
 6. \quad a. \quad \text{Degree of operating leverage} &= \frac{\text{Contribution margin}}{\text{Net operating income}} = \frac{\$300,000}{\$60,000} = 5 \\
 b.
 \end{aligned}$$

Expected increase in sales	8%
Degree of operating leverage	<u>×5</u>
Expected increase in net operating income	<u>40%</u>

- c. If sales increase by 8%, then 21,600 units ($20,000 \times 1.08 = 21,600$) will be sold next year. The new contribution format income statement would be as follows:

	Total	Per Unit	Percent of Sales
Sales (21,600 units)	\$1,296,000	\$60	100%
Variable expenses	972,000	45	75%
Contribution margin	324,000	<u>\$15</u>	<u>25%</u>
Fixed expenses	240,000		
Net operating income	<u>\$ 84,000</u>		

Thus, the \$84,000 expected net operating income for next year represents a 40% increase over the \$60,000 net operating income earned during the current year:

$$\frac{\$84,000 - \$60,000}{\$60,000} = \frac{\$24,000}{\$60,000} = 40\% \text{ increase}$$

Note from the income statement above that the increase in sales from 20,000 to 21,600 units has increased *both* total sales and total variable expenses.

7. a. A 20% increase in sales would result in 24,000 units being sold next year: $20,000 \text{ units} \times 1.20 = 24,000 \text{ units}$.

	Total	Per Unit	Percent of Sales
Sales (24,000 units)	\$1,440,000	\$60	100%
Variable expenses	1,152,000	48*	80%
Contribution margin	288,000	<u>\$12</u>	<u>20%</u>
Fixed expenses	210,000†		
Net operating income	<u>\$ 78,000</u>		

*\$45 + \$3 = \$48; $\$48 \div \$60 = 80\%$.

†\$240,000 – \$30,000 = \$210,000.

Note that the change in per unit variable expenses results in a change in both the per unit contribution margin and the CM ratio.

$$\begin{aligned}
 b. \quad \text{Unit sales to break even} &= \frac{\text{Fixed expenses}}{\text{Unit contribution margin}} \\
 &= \frac{\$210,000}{\$12 \text{ per unit}} = 17,500 \text{ units}
 \end{aligned}$$

$$\begin{aligned}
 \text{Dollar sales to break even} &= \frac{\text{Fixed expenses}}{\text{CM ratio}} \\
 &= \frac{\$210,000}{0.20} = \$1,050,000
 \end{aligned}$$

- c. Yes, based on these data the changes should be made. The changes increase the company's net operating income from the present \$60,000 to \$78,000 per year. Although the changes also result in a higher break-even point (17,500 units as compared to the present 16,000 units), the company's margin of safety actually becomes greater than before:

$$\begin{aligned}\text{Margin of safety in dollars} &= \text{Total sales} - \text{Break-even sales} \\ &= \$1,440,000 - \$1,050,000 = \$390,000\end{aligned}$$

As shown in (5) on the prior page, the company's present margin of safety is only \$240,000. Thus, several benefits will result from the proposed changes.

Glossary

Break-even point The level of sales at which profit is zero. (p. 236)

Contribution margin ratio (CM ratio) A ratio computed by dividing contribution margin by dollar sales. (p. 241)

Cost-volume-profit (CVP) graph A graphical representation of the relationships between an organization's revenues, costs, and profit on the one hand and its sales volume on the other hand. (p. 238)

Degree of operating leverage A measure, at a given level of sales, of how a percentage change in sales will affect profit. The degree of operating leverage is computed by dividing contribution margin by net operating income. (p. 253)

Incremental analysis An analytical approach that focuses only on those costs and revenues that change as a result of a decision. (p. 243)

Margin of safety The excess of budgeted (or actual) dollar sales over the break-even dollar sales. (p. 250)

Operating leverage A measure of how sensitive net operating income is to a given percentage change in dollar sales. (p. 253)

Sales mix The relative proportions in which a company's products are sold. Sales mix is computed by expressing the sales of each product as a percentage of total sales. (p. 256)

Target profit analysis Estimating what sales volume is needed to achieve a specific target profit. (p. 246)

Variable expense ratio A ratio computed by dividing variable expenses by dollar sales (p. 242)